ELEC5206 Laboratory Report

Lab 5 Wind Turbine Power Characteristics

Bench	4	Attendance	Wednesday morning
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1 Aim

The experiment characterised a small wind-turbine generator over three measured wind-speed settings. Load resistance was varied from open circuit to short circuit to obtain the rotor torque-speed, generator current-voltage, mechanical power-speed and electrical power-speed characteristics and identify the maximum power point (MPP).

2 Pre-lab work

1. Wind-speed scaling. For fixed air density and swept area, the available wind power is:

$$P = \tfrac{1}{2}\rho A v^3$$

Doubling wind speed therefore increases the available power by $2^3 = 8$ times, while tripling it increases the power by $3^3 = 27$ times.

2. Rotor efficiency coefficient. The coefficient Cp is the fraction of the wind power intercepted by the blades that is transferred to the rotor as mechanical power. A larger Cp indicates better aerodynamic energy capture. For 500 W of intercepted wind power and Cp = 0.47:

$$P_m = C_p(500\,\mathrm{W}) = 0.47 \times 500\,\mathrm{W} = 235\,\mathrm{W}$$

3. Torque, speed and the MPP. At a fixed wind speed, rotor torque initially rises with rotor speed, reaches a peak, and then falls towards zero at high speed. Because Pm = Tω, maximum mechanical power occurs at an intermediate combination of torque and speed rather than at either torque or speed alone. This is the MPP; operating close to its optimum speed maximises the energy captured as wind conditions change.

3 Experimental method

The generator/controller, variable resistive load and DACI/LVDAC-EMS channels were connected as specified. The load sequence was ∞, 400, 229, 45, 30, 15, 7.5, 5 and 0 Ω. After each operating point stabilised, rotation speed, torque, mechanical power, dc voltage, current and electrical power were recorded. Mechanical and electrical powers were checked using $P_m = 2\pi nT/60$ and $P_l = V_l I_l$. The laboratory sheet prescribed wind speeds of 4, 7 and 10 m/s; however, the first exported data set from this session was labelled and recorded at 5 m/s. The experimental results therefore retain the actual 5, 7 and 10 m/s labels. Negative signs caused by reversed meter polarity are reported as magnitudes only; the measured values were otherwise unchanged.

4 Measured data

Table 1 contains all 27 operating points in the original measurement order. Signs caused solely by reversed meter polarity have been corrected by using absolute values.

Table 1. Wind-turbine measurements for the three wind-speed settings.

Wind (m/s)	R_l (Ω)	n (r/min)	$\lvert T \rvert$ (N·m)	$\lvert P_m \rvert$ (W)	V_l (V)	I_l (A)	P_l (W)
5	∞	**965**	**0.030**	**3.073**	**65.860**	**0.051**	**3.333**
5	400	872	0.123	11.220	58.360	0.105	6.121
5	229	834	0.169	14.720	55.040	0.203	11.180
5	45	533	0.481	26.830	31.180	0.654	20.440
5	30	458	0.634	30.420	26.390	0.854	22.610
5	15	271	0.583	16.540	13.690	0.898	12.250
5	7.5	110	0.407	4.700	3.659	0.464	1.865
5	5	89	0.359	3.323	2.752	0.514	1.415
5	0	44	0.277	1.263	0.045	0.406	0.019
7	∞	**1391**	**0.022**	**3.145**	**95.430**	**0.020**	**1.910**
7	400	1301	0.160	21.740	88.090	0.206	18.140
7	229	1319	0.245	33.890	86.620	0.363	31.550
7	45	889	0.804	74.820	55.370	1.178	65.350
7	30	760	0.921	73.250	44.370	1.456	64.700
7	15	567	1.189	70.620	29.170	1.933	56.430
7	7.5	326	1.141	38.980	13.770	1.813	24.990
7	5	189	0.858	16.980	6.816	1.320	8.970
7	0	64	0.517	3.448	0.115	0.721	0.099
10	∞	**1982**	**0.006**	**1.162**	**136.200**	**0.024**	**3.268**
10	400	1933	0.120	24.210	129.500	0.313	40.600
10	229	1847	0.368	71.110	121.300	0.514	62.480
10	45	1423	1.173	174.700	86.790	1.896	164.900
10	30	1263	1.506	199.100	73.200	2.409	176.600
10	15	1022	2.082	222.700	52.210	3.464	181.000
10	7.5	781	2.648	216.500	32.920	4.339	143.000
10	5	642	2.581	173.500	22.820	4.392	100.300
10	0	128	1.213	16.190	0.291	1.995	0.583

5 Results

5.1 Torque-speed and current-voltage characteristics

Increasing electrical loading reduced generator speed and increased current. At each wind speed, torque first increased as the rotor was loaded and then fell again close to short circuit because the rotor slowed substantially. The dc current-voltage curves have the same broad arch-like behaviour as the torque-speed curves: both quantities rise from the lightly loaded condition, reach a maximum under an intermediate load and decrease toward the heavily loaded endpoint.

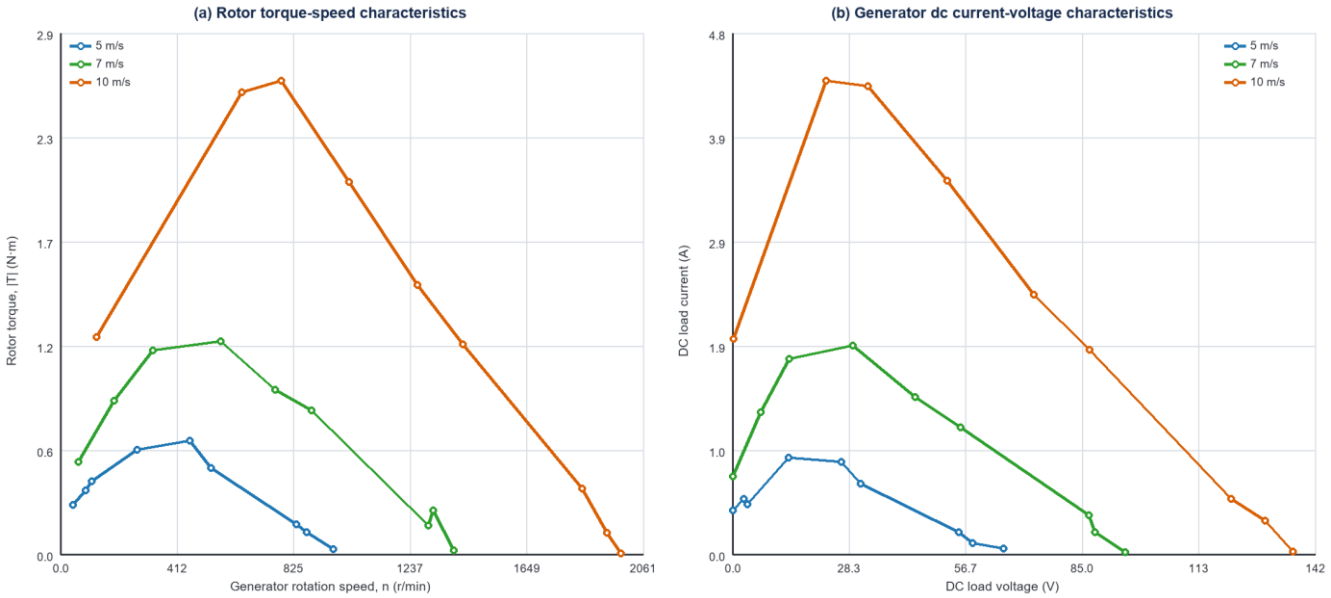


Figure 1. (a) Rotor torque-speed and (b) generator dc current-voltage characteristics at 5, 7 and 10 m/s.

The higher wind-speed curves extend to larger speed, torque, voltage and current. The small non-zero current readings at nominal open circuit arise from measurement offset and limited channel resolution; the measured magnitudes were retained rather than replaced with ideal zero values.

5.2 Mechanical and electrical power characteristics

Both mechanical and electrical power are small at the lightly loaded and near-short-circuit endpoints, and reach a clear maximum at an intermediate rotor speed. The electrical MPP shifts to a lower resistance and much higher speed as wind speed increases.

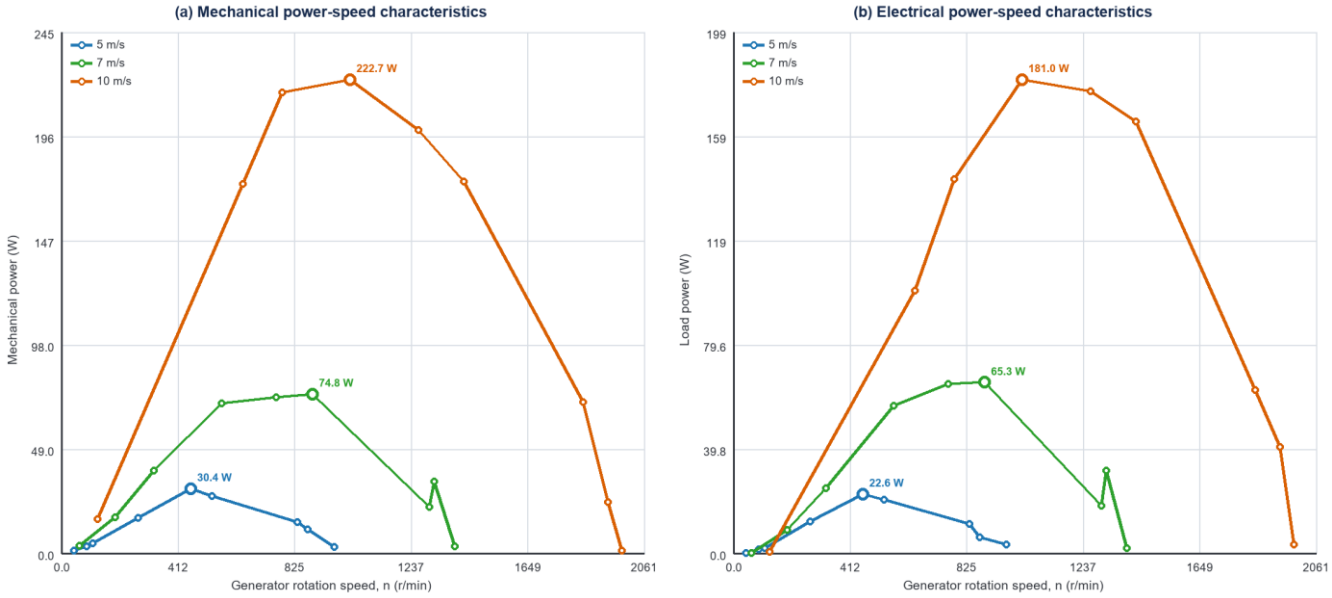


Figure 2. (a) Mechanical and (b) electrical power-speed curves. Open markers identify measured maxima.

Table 2. Measured maximum-power operating points and generator efficiency.

v (m/s)	P_m,max (W)	n at P_m,max (r/min)	|T| at P_m,max (N·m)	P_l,max (W)	n at P_l,max (r/min)	|T| at P_l,max (N·m)	ηg at P_l,max (%)
5	30.42	458	0.634	22.61	458	0.634	74.3
7	74.82	889	0.804	65.35	889	0.804	87.3
10	222.70	1022	2.082	181.00	1022	2.082	81.3

6 Discussion and analysis

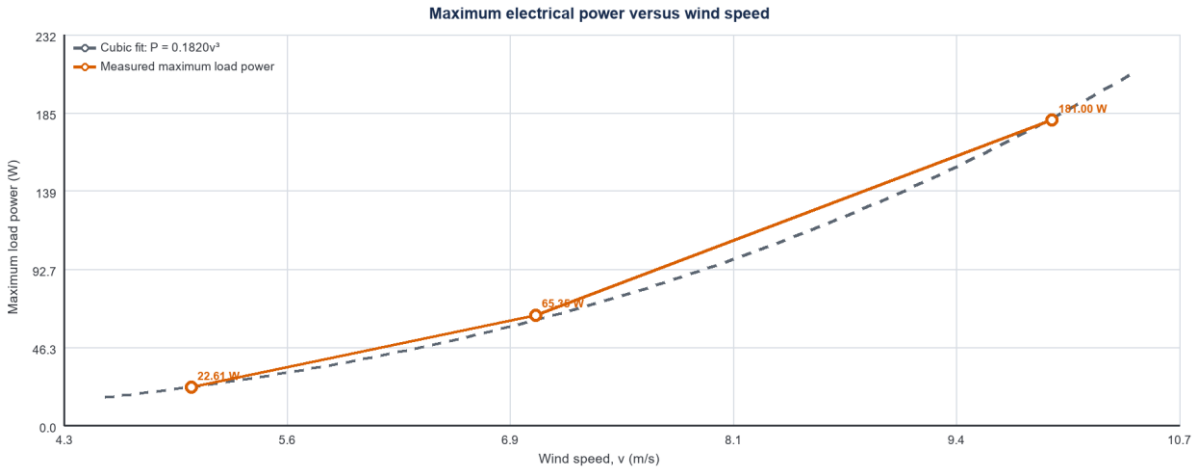


Figure 3. Maximum measured electrical load power and a cubic fit through the origin.

Wind-speed dependence. Maximum electrical power increased from 22.61 W at the measured 5 m/s setting to 65.35 W at 7 m/s and 181.0 W at 10 m/s. Using the 5 m/s point as a cubic-scaling baseline gives 62.04 W at 7 m/s and 180.88 W at 10 m/s. The measured maxima were +5.3% and +0.1% from these estimates. Thus the measured maxima are consistent with the expected cubic trend, although three operating points are not enough to establish an exact power law.

Load matching. At open circuit, voltage and speed were high but current was almost zero. Near short circuit, current remained finite but voltage and speed collapsed. Maximum power therefore occurred between these limits: 30 Ω at 5 m/s, 45 Ω at 7 m/s and 15 Ω at 10 m/s. The optimum resistance is not constant because the generator and turbine operating point changes with wind speed.

Conversion performance. At the electrical MPP, the ratio $P_l/|P_m|$ was 74.3%, 87.3% and 81.3% for 5, 7 and 10 m/s. These are point efficiencies for the generator, rectifier and wiring together, not aerodynamic power coefficients. Values at very low power were not used because small torque offsets can make electrical power appear larger than measured mechanical power.

Uncertainty and improvements. The principal limitations are torque-sensor zero drift, current offset at open circuit, discrete resistor steps, wind-speed tolerance and waiting time before readings stabilise. Repeating each point, zeroing the sensors before every sweep, measuring actual wind speed at the rotor plane and using smaller resistance steps around the peak would improve the MPP estimate.

7 Conclusion

The experiment produced the expected torque-speed, current-voltage and power-speed characteristics of a variable-speed wind turbine generator. Each measured wind speed had a distinct intermediate-load MPP. The maximum electrical powers of 22.61, 65.35 and 181.0 W at 5, 7 and 10 m/s were consistent with cubic wind-speed scaling. The results show why a fixed load cannot maintain maximum power as wind conditions change and why practical systems use converter control to track the optimum operating point.

8 Group contributions

Member	Contribution
Haotian Guo	Data processing, graph preparation and report writing
Yangzheng Liu	Wind-turbine setup, resistor switching and safety checks
Jiayu Yin	LVDAC-EMS recording, measurement checks and result review